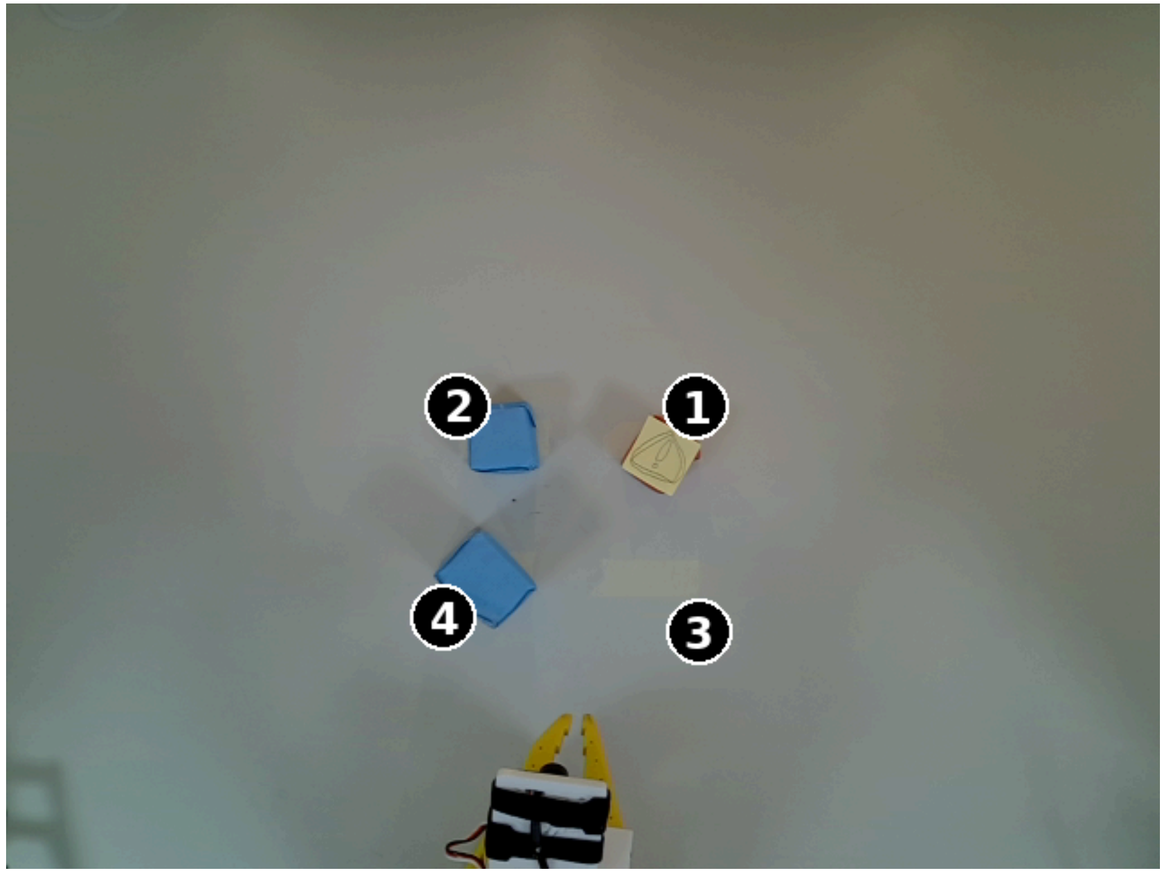
80-SCENE RUN

Scenarios in the 80-scene run

What each scenario tests, and how conformal prediction handles it

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80 scenes · 1,296 instructions · 7 camera sets · scored before and after a wrist look



SETUP

Each split calibrates on 18 scenes and tests on 19

CALIBRATION · 18 SCENES

Only full-name and absent-box instructions in normal light, one row per scene: **18 rows** for Global CP, **54** for the per-count threshold (18 per blocked count).

Output: the threshold. With 18 rows at ε = 0.10 it is the largest calibration score, so it swings between splits: **0.444 to 0.997** for Global CP.

TEST · 19 SCENES

Every instruction type and camera set: **2,261 decisions** per split. Scenes kept out of the split (holdout, duplicate box, physical check, fit, dim, back-light) get the same thresholds.

Output: coverage, set size, actions, wrong grasps. All results in this deck are test results, averaged over **20 random splits**.

37 of the 80 scenes can calibrate or test. Every trial was scored once without and once with the wrist image; each context gets its own threshold.

HOW TO READ THE TABLES

Eight numbers per scenario

COLUMN	MEANING
top-1	share where the model's most probable label is right; no threshold involved
coverage	share where the right label is in the set (target 0.90 at ε = 0.10, 0.80 at ε = 0.20)
set	mean labels in the set, out of 5: spots 1–4 and N for “none”
act · ask · abstain	share of decisions: one box → act, two or more labels → ask, {N} or empty → abstain
wrong grasp	share of all decisions that grasp a box other than the intended one
look first	share the decision policy would send to a wrist look as its first move

Defaults: Global CP (one threshold), ε = 0.10, before the look, test scenes, mean of 20 splits. For a missing box the right label is N; for a vague instruction it is the box the speaker meant.

Ten kinds of instruction

SCENARIO	EXAMPLE	RIGHT ANSWER	INTENDED FIRST MOVE
absent box	“Grasp the small red box”, box removed	none (N)	look, then abstain
no green box	“Grasp the green box”, no green box	none (N)	look, then abstain
no small green box	“Grasp the small green box”	none (N)	look, then abstain
full name	“Grasp the small blue box”	the box named	act; look if hidden
size only	“Grasp the small box”	meant box (2 match)	ask
colour only	“Grasp the red box”	meant box (2 match)	ask
the box	“Grasp the box”	meant box (4 match)	ask
next to	“Grasp the box next to the big blue box”	meant box (2 touch)	ask
left of, robot side	“Grasp the box left of the small red box”	one box	as full name
left of, other side	the same sentence	the speaker's left	ask

“Left of”, “right of” and “in front of” are read in the speaker's frame. Robot side: the speaker stands beside the robot, so both frames agree.

Full-name instructions are the only ones worth acting on

SCENARIO	TOP-1	COVERAGE	SET	ACT	ASK	ABSTAIN	WRONG GRASP	LOOK FIRST
absent box	0.011	0.401	3.48	25.6%	73.9%	0.4%	25.6%	71.0%
no green box	0.978	0.994	3.88	0.1%	82.5%	17.4%	0.1%	99.9%
no small green box	0.989	0.997	3.34	0.0%	77.1%	22.9%	0.0%	100.0%
full name	**0.851**	0.979	3.46	19.2%	80.2%	0.6%	1.5%	75.4%
size only	0.502	0.933	3.92	8.1%	89.3%	2.7%	2.9%	81.1%
colour only	0.480	0.927	3.49	13.6%	85.5%	0.9%	6.3%	68.4%
the box	0.245	0.880	4.26	6.2%	89.8%	4.1%	4.4%	81.7%
next to	0.469	0.920	3.92	5.2%	91.0%	3.8%	2.3%	83.7%
left of, robot side	0.476	0.912	3.84	8.9%	88.6%	2.5%	3.1%	86.0%
left of, other side	0.394	0.895	3.91	8.3%	88.9%	2.9%	4.2%	86.8%
all	**0.587**	**0.930**	**3.74**	**10.1%**	**85.6%**	**4.3%**	**3.3%**	**82.1%**

A missing box is the most dangerous case

25.6%

of decisions grasp some other box when the named box is missing (ε = 0.10)

84.9%

at ε = 0.20

The right answer, N, is in the set only **40.1%** of the time. N is the model's top answer 1.1% of the time before the look and **0.0%** after it.

The average for missing-box instructions looks fine, 0.956, because 266 of its 296 test rows per split are “green” sentences, which the model handles (0.994, 0.997).

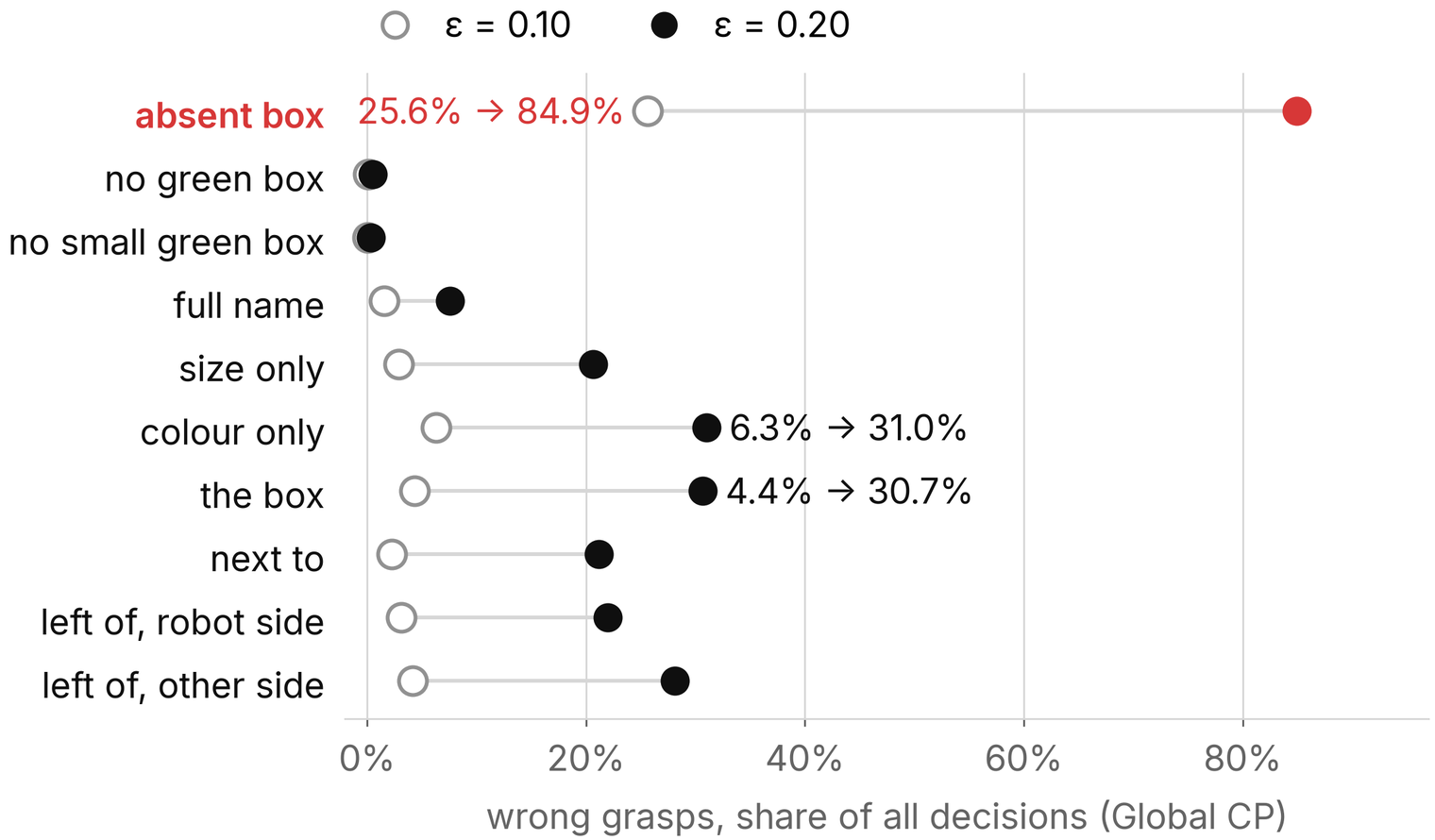
The robot asks on 73.9% of them, about sets of 3.48 labels that usually leave out “none”.

For vague instructions, loosening ε is unsafe

At ε = 0.10 the robot asks 86–91% of the time on vague instructions, which is the intended move.

At ε = 0.20 it grasps 40–62% of the time on them, and 21–31% of all their decisions grasp the wrong box.

For “the box”, top-1 is 0.245: chance with four candidates.

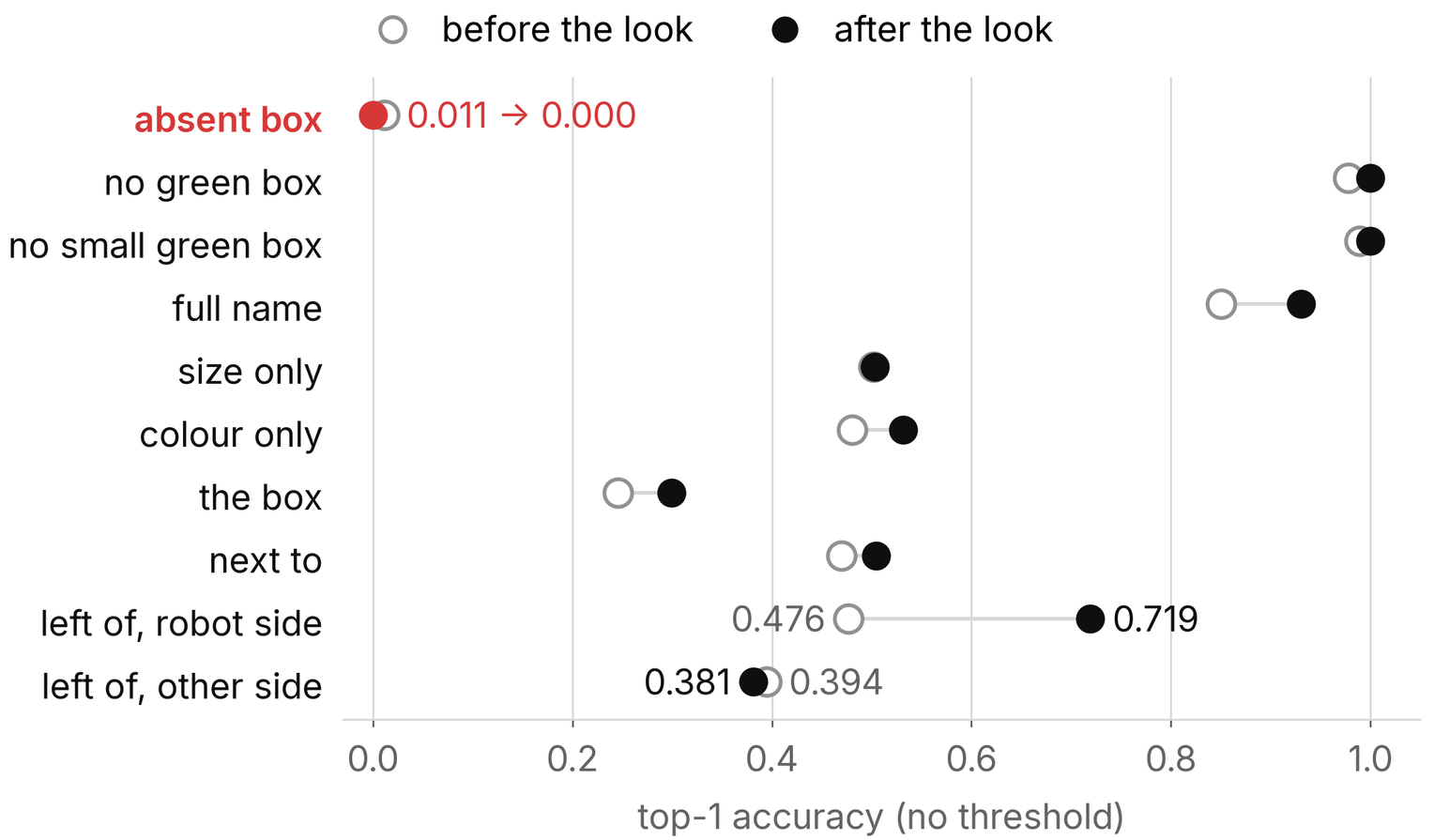


The look helps only where the image holds the answer

Left of, robot side:
0.476 → 0.719.
Full name: 0.851 → 0.931.

Left of, other side:
0.394 → 0.381. No image shows where the speaker stands.

Absent box: 0.011 → 0.000.



Seven camera sets, blocked in software

CAMERAS	BLOCKED	WHAT THE MODEL SEES
top + left + right	0	everything the fixed cameras see
top + left	1	top view and one side view
top + right	1	top view and the other side view
left + right	1	both side views, no top view
top only	2	colour and position; big and small look alike from above, and one box's top carries a sticker
left only	2	one side view: colour and size
right only	2	the other side view: colour and size
none	3	no image: handled by rule (all four boxes, then look), not run

Blocking a camera means leaving its image out of the prompt. Every scene appears in all seven sets.

The top camera alone is the weakest view

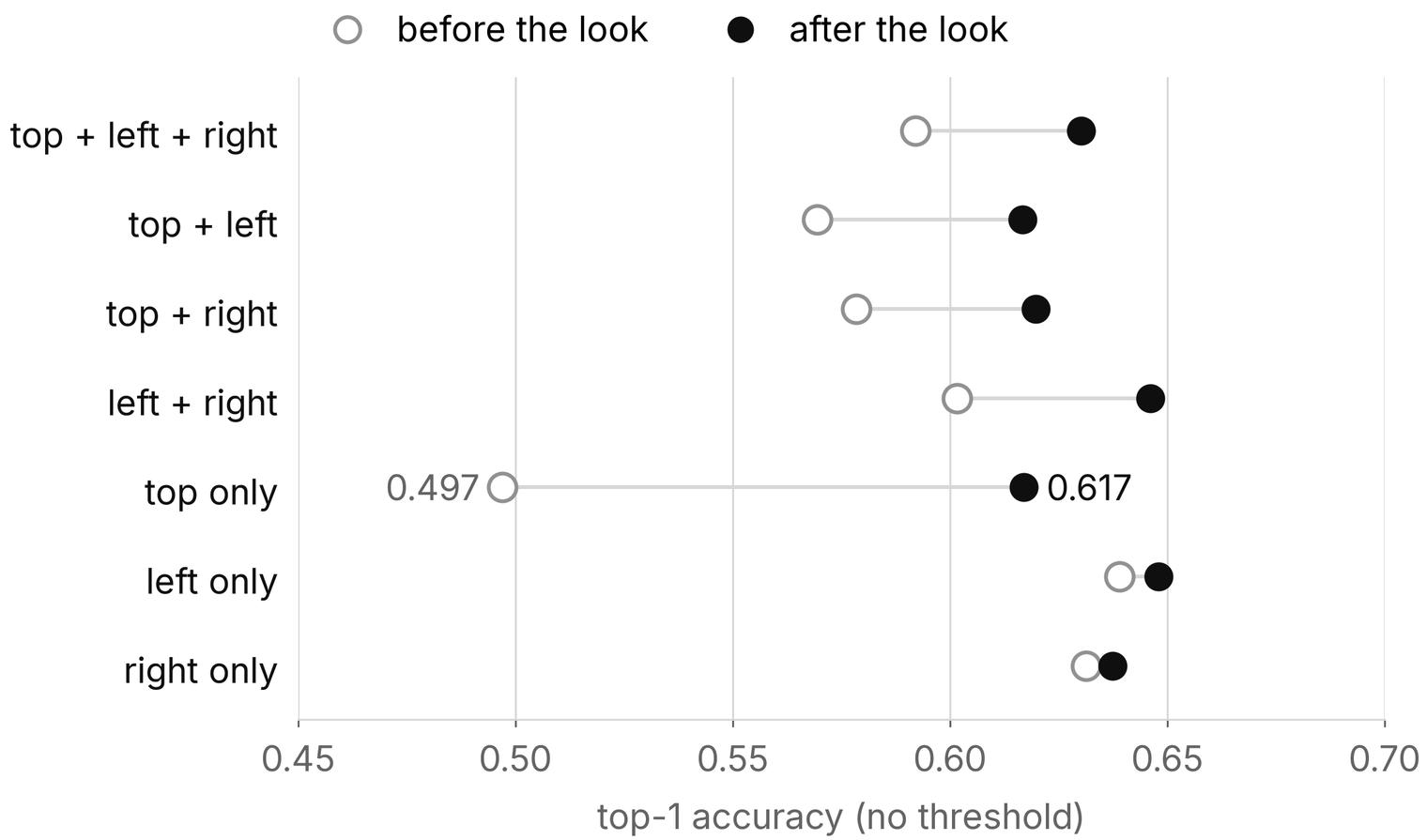
		GLOBAL CP					PER COUNT			
CAMERAS	TOP-1	COVERAGE	SET	ACT	WRONG	LOOK FIRST	COVERAGE	SET	ACT	WRONG
top + left + right	0.592	0.936	3.98	7.7%	2.5%	64.5%	0.913	3.77	15.5%	6.5%
top + left	0.570	0.930	3.83	8.8%	3.0%	87.2%	0.968	3.94	5.7%	1.6%
top + right	0.578	0.934	3.83	8.7%	2.9%	87.3%	0.971	3.95	5.2%	1.3%
left + right	0.602	0.939	3.92	8.5%	2.3%	87.8%	0.974	4.03	5.3%	1.2%
top only	**0.497**	0.908	3.53	10.3%	**4.6%**	84.8%	0.964	3.97	4.1%	1.8%
left only	0.639	0.928	3.53	13.4%	3.8%	81.2%	0.973	3.94	6.0%	1.2%
right only	0.631	0.933	3.56	13.5%	3.7%	81.4%	0.977	3.98	5.8%	0.7%

All instruction types. “Per count” is one threshold per number of blocked cameras. Adding the top camera to side views lowers top-1 in every pairing.

The look repairs the top camera

Top only: top-1 **0.497 → 0.617** after the look, the largest gain; all sets then sit at 0.617–0.648.

At ε = 0.20 without a look, top only grasps the wrong box in 25.0% of decisions; a per-count threshold cuts that to 13.3% by grasping less.



The look raises accuracy, grasps and wrong grasps

	TOP-1	COVERAGE	SET	ACT	ASK	ABSTAIN	WRONG GRASP
Global, before	0.587	0.930	3.74	10.1%	85.6%	4.3%	3.3%
Global, after	**0.631**	0.908	3.66	16.3%	78.0%	5.6%	5.6%
per count, before	0.587	0.963	3.94	6.8%	91.4%	1.8%	2.0%
per count, after	**0.631**	0.910	3.62	13.5%	79.5%	7.0%	4.2%

“After the look” means every trial was scored again with the wrist image added, under its own threshold. No rule decided to look. ε = 0.10.

Eight kinds of scene

POOL	SCENES	WHAT IT IS	ROLE
standard	32 (27 used)	standard scenes, all four boxes	calibrate and test
absent box	12 (10 used)	one box removed	calibrate and test
holdout	6 evaluated	arrangements never used for calibration	test only
duplicate box	8	two identical boxes; no “the box”, “next to” or “left of” instructions	descriptive
physical check	4	reserved for a camera-cover check; no cover used	descriptive
fit	8	used only to fit the temperature	descriptive
dim / back-light	8 / 8	lighting conditions cancelled by auto-exposure	not a lighting result

One holdout scene is excluded: its scene card contradicts the photos.

Holdout scenes cover slightly less

POOL	TOP-1	COVERAGE	SET	ACT	WRONG GRASP	ε 0.20 COVERAGE	ε 0.20 WRONG
standard	0.561	0.932	3.76	9.0%	3.1%	0.673	20.0%
absent box	0.675	0.920	3.67	13.5%	3.8%	0.739	18.2%
holdout	0.528	0.918	3.77	9.8%	3.8%	0.645	23.1%
duplicate box	0.704	0.956	3.60	13.5%	2.2%	0.791	12.9%
physical check	0.618	0.943	3.80	8.9%	2.3%	0.719	16.3%
fit	0.565	0.935	3.80	8.6%	3.0%	0.678	20.7%
dim	0.585	0.938	3.82	8.7%	2.8%	0.693	18.8%
back-light	0.559	0.927	3.78	9.4%	3.4%	0.660	21.3%

Pools differ in instruction mix: duplicate-box scenes carry no “the box”, “next to” or “left of” instructions, and absent-box scenes have one distractor fewer. Dim and back-light match normal light because the change never reached the images.

SUMMARY

Act on full names; a missing box is the worst case

1. Full-name instructions are the only ones where grasping pays: top-1 0.851, wrong grasps 1.5%.
2. A missing box is the most dangerous case: 25.6% wrong grasps at ε = 0.10, 84.9% at ε = 0.20.
3. Vague instructions should be asked about; loosening ε turns them into 21–31% wrong grasps.
4. The top camera alone is the weakest view, and the wrist look repairs it (0.497 → 0.617).

NEXT STEPS

- Find why the sets for an absent box lack “none”.
- Rerun the lighting pools with auto-exposure off.
- Correct the card of the excluded holdout scene.